

Sentiment Analysis on Restaurant Reviews-

Comparing three embedding methods Bag-of-words vs Word2Vec vs DistilBERT

About project

The project mainly deals with sentiment analysis of Restaurant Review which consists of 1000 pre-defined labelled dataset which is either positive or negative. The core idea is about to compare the three embeddings namely Bag-of-Words, Word2Vec (it is considered to be trained from the corpus), DistilBERT and these three embeddings are paired with three classifier models to compare namely Navie Bayes, Long Short Term Memory (LSTM) and feed-forward network. These models run through similar shared pipeline due to which the representation and the classifier changes. These 3 embeddings and 3 models combinely form 9 combinations total, scored and laid out in single accuracy grid.

Methodology

The dataset is about 1000 restaurant reviews which is of tsv file, where positive and negative feedback is evenly split among the data. So, there is no need to deal with class imbalance. Beginning with data cleaning such that it strips the unnecessary tokens which isn't letter and drop stopwords. By holding the words like "not" because if it is removed it can reverse the meaning of the review, such as changing negative to positive for example "not good" to "good". Then followed by stem what's left with a porter stemmer.

From there, three embeddings turn the cleaned text into vectors:

- Bag-of-Words — word counts capped at 1500 features, no notion of order or meaning
- Word2Vec — trained from scratch on the corpus itself, 100 dimensions, reviews represented as the average of their word vectors
- DistilBERT — pretrained transformer, used frozen (no fine-tuning), mean-pooled hidden states as the document vector

And three classifiers sit on top of whichever embedding is active:

- Naive Bayes (Gaussian, on scaled features)
- a small LSTM
- a feed-forward network

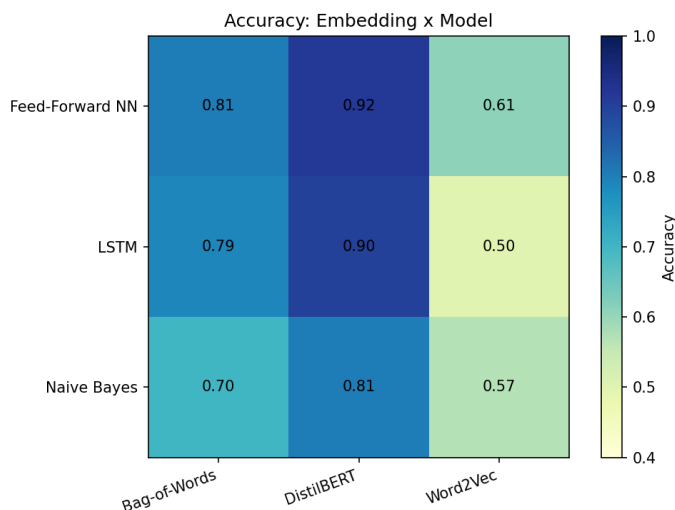
Tough Phase

I could say this is the most time-consuming phase where at the beginning of this pipeline gave me a results grid where Bag-of-Words and DistilBERT were achieved accuracy of 100% which is considered to be unachievable. This is because not calling `train_test_split` properly. When I was sorting the data frame and slicing based on row position, so due to some reviews which are actually got into both training and test datasets. There are few similar reviews and since bag-of-words and DistilBERT both are nearly identical text to nearly identical vectors, such that the models were just recognizing text that can memorized. Based on Word2Vec had noticed the issue because it should a different increase compare to remaining methods, as its vectors are well trained from scratch due

to that they may be too noisy to memorize the leaked data. To fix this problem, I had properly stratified `train_test_split` and also my making every embedding fits only to the training data rather than using the full data frame. In result, the accuracy dropped to actual numbers which are presented in the result.

Result

The actual output has been presented in the form of confusion matrix,



The combination of DistilBERT with feed-forward neural network has the highest accuracy rate that is 92%. By observing the confusion matrix, we can say that DistilBERT was concluded as the strongest embedding across all the classifier that is around 81% to 92%. Followed by Bag-of-Words considered as a solid, cheap baseline consists of accuracy between 70% to 81%. Whereas, Word2Vec was considered as weakest among embedding based on the accuracy that is between 50% to 61%. As mentioned, when Word2Vec is combined with LSTM considered to achieve lowest accuracy among all that is 50% that mentioned the binary label. Based on output achieved the training word vectors from the scratch needs more training reviews to produce anything meaningful insight, as 100 dimensions spread across a vocabulary where signal isn't enough but where as DistilBERT is considered to be pretrained on a huge amount of text and it doesn't require the dataset to learn the words which are presented at the beginning of the review.

Conclusion

Once dataset is fixed, the DistilBERT which is considered to be pretrained transformer beats both the remaining embedding methods namely from scratch embedding (Word2Vec) and Bag-of-Words. Though we can also consider that Bag-of-Words isn't consider as a bad choice, specifically in the case of something need to be achieved fast when it doesn't require any GPU. Word2Vec has low accuracy compare to other because of learning everything about the words from the scratch. Whereas, DistilBERT didn't have that problem since it already knows how language works from being trained on a massive amount of text beforehand.